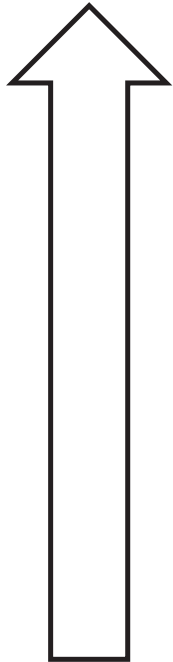




8.6.4 Wrap-Up: Reflection

1. Color in the arrow up to the statement which best describes your current understanding.



I'm so confident, I could explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard, and I listened, but I am finding this challenging. I will make sure that I get help.

I do not understand any of this yet. I listed my questions below.

Unit 8, Lesson 7: Completing the Square Algebraically



Benchmarks

MA.912.AR.3.6 Given an expression or equation representing a quadratic function, determine the vertex and zeros, and interpret them in terms of a real-world context.

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.



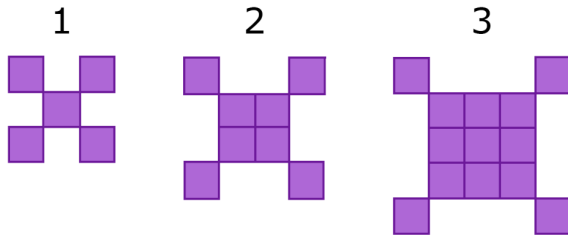
Learning Targets

- ☐ I can complete the square for a quadratic expression algebraically.
- ☐ I can rewrite a quadratic equation in vertex form by completing the square.



8.7.1 Warm-Up: Square Pattern

Three steps of a visual pattern are shown.



1. Use the pattern to complete the table.

n	Calculation	# of squares
1		
2	$2^2 + 4$	
3		
4		
43		
n		

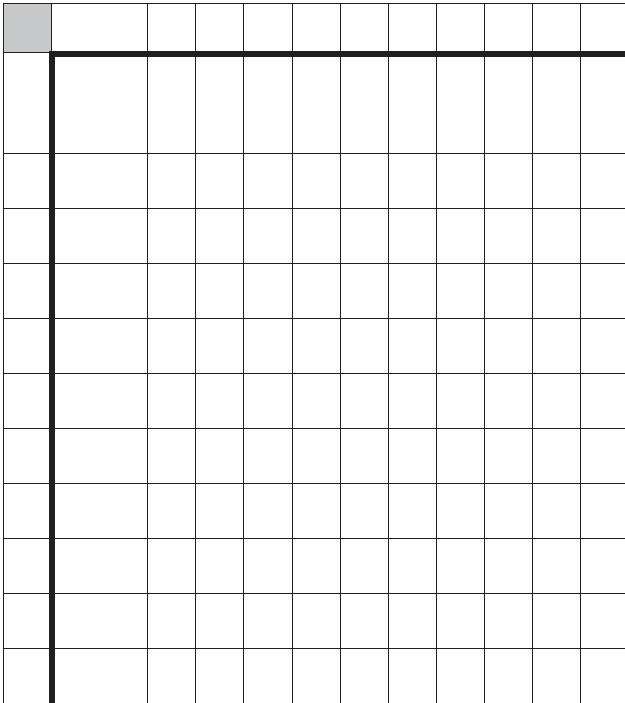


8.7.2 Collaboration: Completing the Square Algebraically

For each expression, the first step to completing the square is shown.

Expression A	Expression B	Expression C
$6x^2 - 24x + 15$	$x^2 - 24x + 15$	$4x^2 - 24x + 15$
$6(x^2 - 4x) + 15$	$(x^2 - 24x) + 15$	$4(x^2 - 6x) + 15$

1. Discuss with your partner how to determine the value to add to the expression inside the parentheses that results in a perfect square trinomial. Use the algebra tiles diagram provided if sketching a model will help you.



2. Use your process to complete each perfect square trinomial.

Expression A	$6(x^2 - 4x + \underline{\hspace{1cm}}) + 15$
Expression B	$(x^2 - 24x + \underline{\hspace{1cm}}) + 15$
Expression C	$4(x^2 - 6x + \underline{\hspace{1cm}}) + 15$

3. In Expression A, the value that was added to create the perfect square trinomial is being multiplied by _____, the coefficient of the perfect square trinomial. The value that should be subtracted from 15 to maintain equivalence is _____.
4. In Expression B, the value that was added to create the perfect square trinomial is being multiplied by _____, so the value that should be subtracted from 15 to maintain equivalence is _____.
5. In Expression C, the value that was added to create the perfect square trinomial is being multiplied by _____, the coefficient of the perfect square trinomial. The value that should be subtracted from 15 to maintain equivalence is _____.

6. Factor each perfect square trinomial and rewrite as a binomial squared. Also, enter the value determined from the steps described above, including the appropriate sign (+ or -).

Expression A	$6(x - \underline{\hspace{1cm}})^2 + 15 \underline{\hspace{1cm}} \underline{\hspace{1cm}}$
Expression B	$(x - \underline{\hspace{1cm}})^2 + 15 \underline{\hspace{1cm}} \underline{\hspace{1cm}}$
Expression C	$4(x - \underline{\hspace{1cm}})^2 + 15 \underline{\hspace{1cm}} \underline{\hspace{1cm}}$

Rewriting the resulting perfect square trinomial as a binomial squared is the next step in the process, as it represents the factored form of a perfect square trinomial. The algebra tiles are a visual representation of this perfect square.

$$x^2 + 8x + 16 = (x + 4)^2$$

		$x + 4$			
x	x^2	x	x	x	x
+					
4	x	1	1	1	1
	x	1	1	1	1
	x	1	1	1	1
	x	1	1	1	1

The process of completing the square is used to rewrite quadratic expressions, equations, or functions from standard form, $ax^2 + bx + c$, to vertex form, $a(x - h)^2 + k$.

This process is helpful when trying to find key features that are not easy to determine using standard form.

7. While learning the process of completing the square with algebra tiles, the leading coefficients were all positive. The table shown gives a quadratic equation in standard form that has negative coefficients.

Complete the table to complete the square without using algebra tiles.

Process	Equation
Original equation	$y = -3x^2 - 24x + 11$
a. Factor out the common factor of the first two terms.	$y = \underline{\hspace{1cm}}(x^2 + \underline{\hspace{1cm}}x) + 11$
b. Discuss with your partner why there is a plus sign instead of a minus sign in the parentheses. $(x^2 \oplus \underline{\hspace{1cm}}x)$	
c. Complete the statement. There is a plus sign instead of a minus sign in the parentheses because ... _____.	
d. Determine the value that makes a perfect square trinomial in the parentheses.	$y = \underline{\hspace{1cm}}(x^2 + \underline{\hspace{1cm}}x + \underline{\hspace{1cm}}) + 11$
e. Then, to maintain equivalence, multiply the factor by the value . Add the opposite of that value .	$y = \underline{\hspace{1cm}}(x^2 + \underline{\hspace{1cm}}x + \underline{\hspace{1cm}}) + 11 + \underline{\hspace{1cm}}$

- f. Discuss with your partner why, after finding the product of the **factor** and the **value** added to make a perfect square trinomial, the opposite of this product is added to the expression. Summarize your discussion.

- g. Rewrite the perfect square trinomial represented in the parentheses as a binomial squared.

$$y = \underline{\hspace{1cm}}(x + \underline{\hspace{1cm}})^2 + 11 + \underline{\hspace{1cm}}$$

- h. Combine like terms.

$$y = \underline{\hspace{1cm}}(x + \underline{\hspace{1cm}})^2 \underline{\hspace{1cm}} \underline{\hspace{1cm}}$$



8.7.3 Guided Practice: Completing the Square Algebraically

Rewrite each equation in vertex form by completing the square. Then, give the key features revealed by both equations.

1. $y = x^2 + 22x - 53$

Key Features					
x-intercept		vertex		domain	
x-intercept		increasing		range	
y-intercept		decreasing		opens	
left end behavior	$x \rightarrow -\infty, y \rightarrow \underline{\hspace{2cm}}$				
right end behavior	$x \rightarrow \infty, y \rightarrow \underline{\hspace{2cm}}$				

2. $f(x) = -8x^2 + 80x - 71$

Key Features					
x-intercept		vertex		domain	
x-intercept		increasing		range	
y-intercept		decreasing		opens	
left end behavior	$x \rightarrow -\infty, y \rightarrow \underline{\hspace{2cm}}$				
right end behavior	$x \rightarrow \infty, y \rightarrow \underline{\hspace{2cm}}$				



8.7.4 Wrap-Up: Cool Down

1. Rewrite the equation in vertex form by completing the square. Then, give the key features revealed by both standard form and vertex form.

$$y = -2x^2 + 12x + 1$$

Key Features

x-intercept		vertex		domain	
x-intercept		increasing		range	
y-intercept		decreasing		opens	
left end behavior	$x \rightarrow -\infty, y \rightarrow \underline{\hspace{2cm}}$				
right end behavior	$x \rightarrow \infty, y \rightarrow \underline{\hspace{2cm}}$				



Unit 8, Lesson 8: Modeling With Quadratics in the Real World



Benchmarks

MA.912.AR.3.6 Given an expression or equation representing a quadratic function, determine the vertex and zeros and interpret them in terms of a real-world context.

MA.912.AR.3.8 Solve and graph mathematical and real-world problems that are modeled with quadratic functions. Interpret key features and determine constraints in terms of the context.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.



Learning Targets

- ☐ I can interpret key features of quadratic relationships in a real-world context.
- ☐ I can interpret solutions as viable or non-viable options given a real-world context.
- ☐ I can represent constraints of a real-world context using set-builder notation.